The summand-swap invariant does not lower the closure degree of the chained S_3 descent

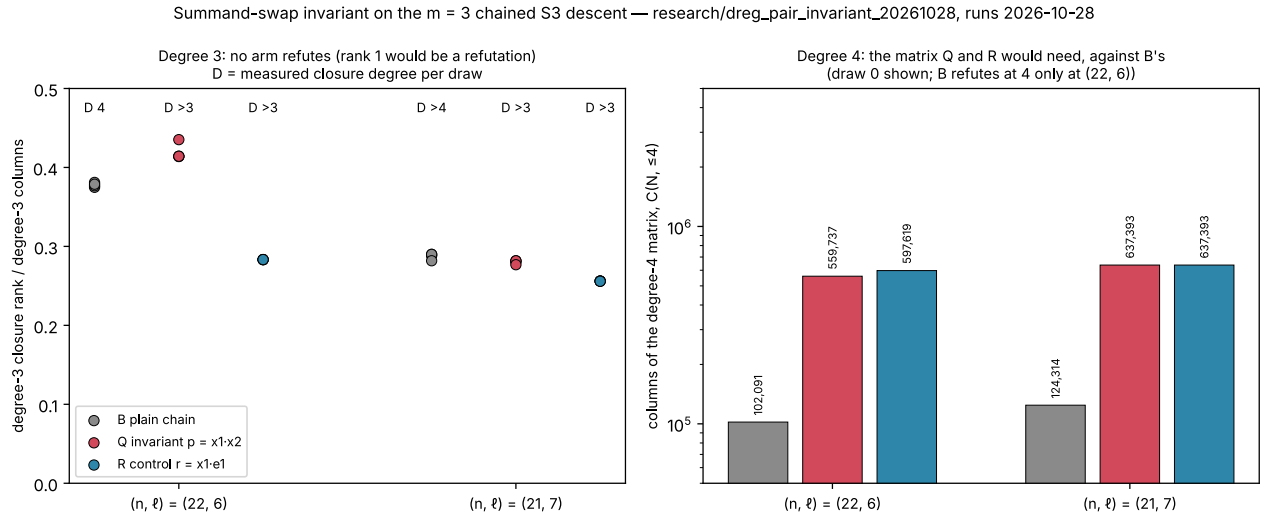
Study `research/dreg_pair_invariant_20261028/`, pre-registered. **Class:** stage diagnostic. **Verdict:** H_none in both cells.

Question

At m = 3 every cubic monomial of the chained descent comes from $x_1x_2e_1$. In the swap invariants $s = x_1+x_2$ and $p = x_1x_2$, the first link becomes the quadratic $s^2e_1^2 + p\,e_1 + p^2 + b$.

Does adding p as unknowns let the MutantXL closure refute at a lower degree? Is any gain due to the invariant, or only to quadratisation? Quadratisation alone is tested by a matched control, $r = x_1e_1$.

Result



cell	B (plain)	Q (invariant)	R (control)
(22, 6), 4 draws	refutes at 4	> 3	> 3
(21, 7), 4 draws	> 4	> 3	> 3

- Q's degree 4 is unmeasured. Its dense matrix needs 523,686–637,393 columns, against B's 102,091 and 124,314.
- So at (22, 6) the invariant arm refutes no earlier than B, at 5.1–5.5 times the columns.
- Quadratisation alone (R) does not help either.
- Controls valid: exact identity check, B reproduces its records, and no satisfiable draw is refuted.

Reading

- **This closes the pair-invariant lever for the chain at m = 3.** At m = 4 the second cubic link has no summand symmetry and would need the R construction, which also did not help.
- **Unregistered pointer:** at (22, 6), Q closes more of its degree-3 matrix (0.41–0.44) than B (0.37–0.38) or R (0.28), but not enough to refute.
- **No row moves and none beats rho.** This is a negative against the success criterion.
- **Untried successors:** the invariants as a replacement for the summand coordinates, so that N falls; the orbit-aware split rule.